

Smart Home Management System

Abstract

The Smart Home Management System is an integrated solution designed to automate, monitor, and control household devices efficiently.

The system improves home security, energy consumption, and user comfort by providing intelligent control over lighting, temperature, doors, appliances, and renewable energy source.

It is implemented using Java(OOP principles), JavaFX for GUI, and SQL Server as the backend database.

This Project introduces a flexible user system with personalized profiles, access control levels, device management, and energy-saving insights to ensure an efficient smart home environment.

Introduction

The rapid growth of smart technologies has transformed traditional home into intelligent spaces capable of responding to user's needs.

Smart home aim to make living more comfortable, efficient, and secure by automating daily tasks such as controlling lights, doors, refrigerators, and more.

This Project presents a complete and scalable Smart Home Management System, allowing users to monitor and control their home devices through a simple and intuitive graphical user interface (GUI).

Problem Statement

Managing a modern home with multiple device is becoming more complex.

Users often face challenges such as :

- Wasting energy due to manual device control.
- Lack of monitoring over device usage.

- Difficulty managing access permissions for different family members.
- No unified system to control lighting, temperature, appliances ,and energy units.

This project solves this issues by offering a centralized smart home platform.

A unified system is required to monitor, control, and manage hoem devices in an organized, intelligent manner.

System objectives

The system aims to:

- Provide an easy-to-use interface for controlling smart home devices.
- Improve home security through user management and access control.
- Reduce energy consumption using energy-saving modes.
- Give real-time feedback on device status.
- Support multiple user roles(Admin-Parent-Kid-Guest).
- Show clear analytics and usage statistics.
- Offer a personalized user experience (Themes, Preferred modes, notifications.....).

Project Scope

In-Scope

- Smart home device modeling(AC, Lights, Door, Refrigerator, Solar Panel, Wind Turbine)
- User account system with roles and permissions.
- Login system using JavaFX.
- Real-time device control dashboard
- SQL Server database integration.
- Device usage statistics& energy tracking
- MVC architecture implementation.

Out of scope

- Mobile application
- IoT physical device (hardware)
- voice control system

System Features

User Management

- Register/ Login
- Two-Factor Authentication
- Theme & Language setting
- User profile
- User roles (Admin-Parent-Kid-Guest).

Device Control

- ON/OFF
- Change AC temperature
- Adjust light brightness
- Lock/Unlock door
- Refrigerator controls
- Renewable energy monitoring

Analytics & Monitoring

- Total usage hours
- Last used device
- Energy saving mode
- System activity logs

Admin panel

- Manage user

- Assign permissions
- Add/remove rooms or devices
- View system logs

System Architecture

The system uses the MVC architecture:

Model

contains all smart home classes

- Device (Abstract)
- Light
- Door
- Refrigerator
- AC
- Renewable Energy Source (Abstract)
- Solar Panel
- Wind Turbine
- Room
- User

view

JavaFX UI:

- loginView
- DashboardView
- DeviceControlPanels

Controller

- HomeController
- LoginController

- DeviceController

Database Layer

- DBConfig
- DatabaseManager

Implementation Details

Application Environment

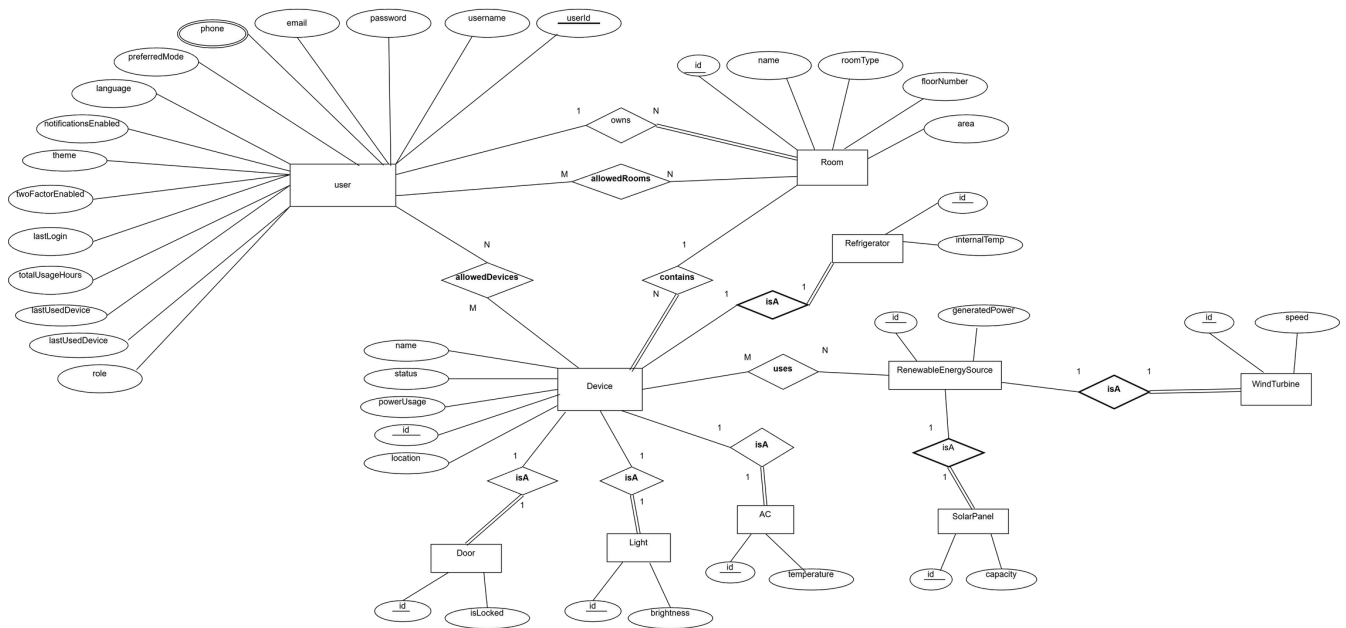
- Desktop Application
- Technology: JavaFX, FXML, SQLServer
- IDE: IntelliJ
- Design Tool: Scene Builder
- Database Tool: SQL Server Management Studio

Database Structure (SQL Server)

Tables include:

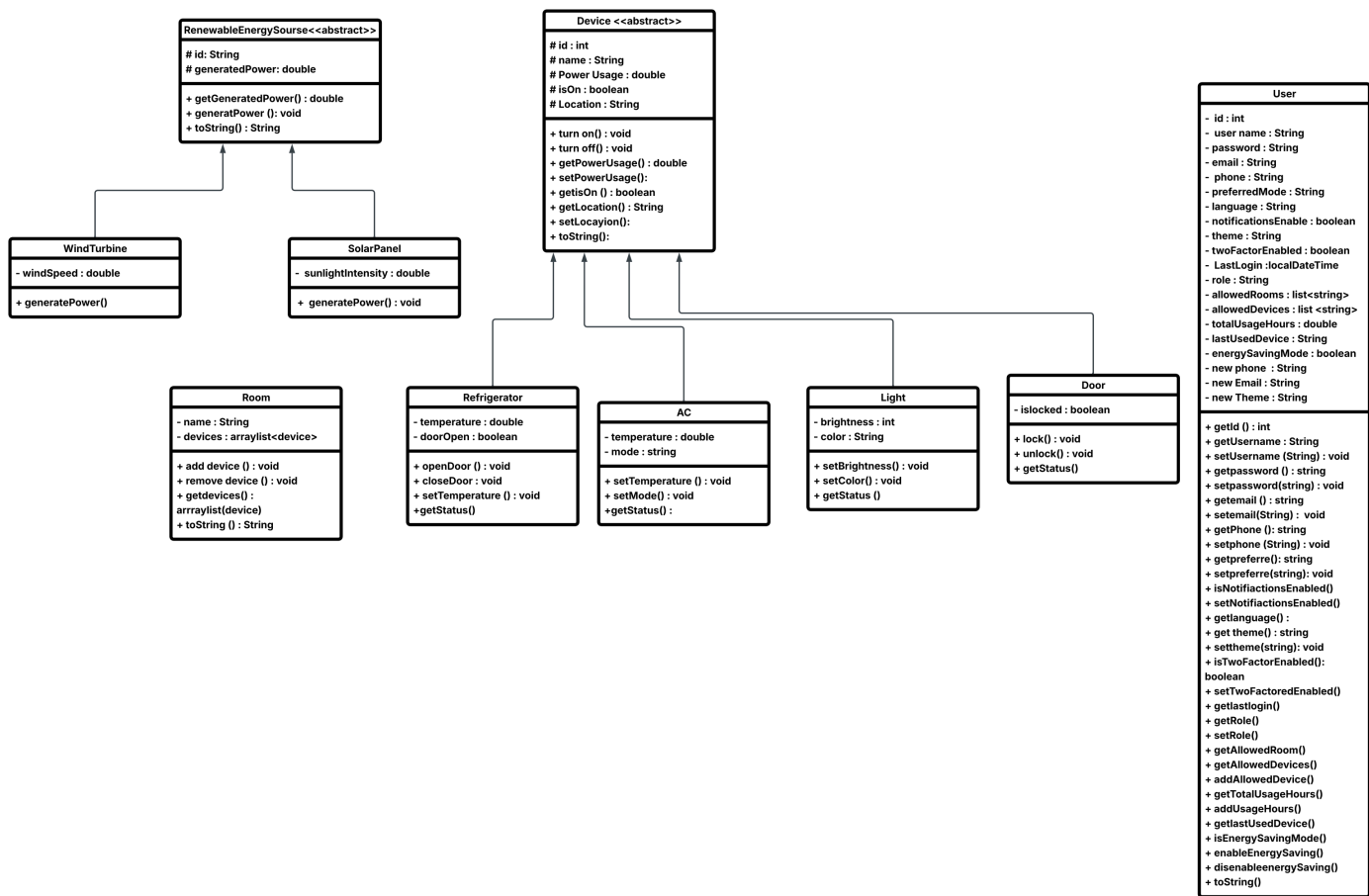
- Users
- Rooms
- Device
- DeviceStatus
- EnergyUsage
- Logs

The ERD describes relationships between users, room, devices, and usage logs.(Designed by Ola Ahmed)



UML Diagrams Included

- Use Case Diagram
- Class Diagram
- Sequence Diagram
- They describe the full workflow and system structure. (Designed by Menna AbdElkreem)



Project Modules

Login Module

Allows users to authenticate and access the system based on their roles.

Dashboard Module

Displays device status, statics, and quick controls.

Device Management Module

Allows user to view and control all smart devices in each room.

User Administration Modul

Admin-only features:

Add User

Remove User

Edit permissions

Testing Plan

- Unit testing for each class
- GUI testing
- Integration testing
- Database connection testing
- Scenario testing(Home, Away, Night, modes)

Expected Results

- A function desktop smart home application
- clean, styled, modern UI
- Smooth login experience
- Real-time device control
- Energy Usage insights
- Professional UML + ERD + documentation

Conclusion

This proposal outlines a complete Smart Home Management System with robust features, modular architecture, and good scalability.

The project demonstrates strong understanding of OOP, JavaFX, SQL server.

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